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PATENTS AND PATENTEES IN THE DUTCH REPUBLIC BETWEEN c.1580 AND 1720

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Abstract: This paper examines the working of the patent system in the Dutch Republic during the period when the incidence of patenting reached its peak, between around 1580 and 1720. It looks at the system primarily from the viewpoint of the patent-holders, by analyzing the geographical background and social profile of patentees, the evolution of careers of patenting and the growth of commercial tendencies in the obtainment and exploitation of patents. It is argued that the development in the Netherlands in these respects resembled the evolution in England rather than that in France or the United States, but that in contrast with England the practice of patenting in the Dutch Republic to some extent fell into disuse. A turning point was reached around 1640. While the number of patents granted by the States General and the States of Holland began a long-term decline and the filing of specifications became a more and more rare occurrence, the relative importance of "multiple" patentees diminished, the geographical spread of patentees lessened and the role of the commercial and professional classes increased. Although a decrease in the number of patents did not necessarily reflect a falling-off in the rate of invention, it certainly meant a decline in the demand for patents on the side of inventors. The author suggests that this decline must at least in part have been induced by the evolution of the patent system itself.

INTRODUCTION

Since the publication of studies by G. Doorman and M. Silberstein it is well-known that the Dutch Republic possessed a highly developed system of patents for invention.¹ Silberstein even went so far as to suggest that, thanks to the growing influence of Dutch institutions in the seventeenth century in general, the practice of patenting in the United Provinces in turn stimulated the emergence of patent systems in other parts of Europe.²

This paper will examine the working of the patent system in the Dutch Republic during the period when the incidence of patenting reached its peak, between around 1580 and 1720.³ In common with recent innovative studies on the patent systems in England and France between c.660 and 1800 and the United States between c.790 and 1850 it will look at the Dutch system primarily from the viewpoint of patent-holders, by analyzing the geographical background and social profile of patentees, the evolution of careers of patenting and the growth of commercial tendencies in the obtainment and exploitation of patents.⁴ It will be shown that the pattern of development in the Netherlands in these respects resembled the evolution in England rather than that in France or the United States, but that in contrast with England the practice of patenting in the Dutch Republic to some extent fell into disuse, due to factors that at least in part must have resided within the patent system itself.

THE DEVELOPMENT OF THE PATENT SYSTEM IN THE DUTCH REPUBLIC

Patents for invention in the Dutch Republic were granted by three different authorities: the central governing body of the state (the States General), the estates of the separate provinces which jointly formed the Republic, and town governments. Between c.1590 and the middle of the seventeenth century, the largest number of patents was issued by the States General in The Hague. After that, the States General ceded place as the chief patent-issuing body to the estates of the dominant province of the Republic, Holland, as Table 1 shows. This shift of primacy occurred hardly a decade after the States General had made it a rule to prescribe that an invention could only be brought into practice in a particular province of the Republic if the patent had been formally confirmed by the estates of the province concerned. It almost exactly coincided with a similar shift from the States General to the States of Holland in the issue of copyrights on books.⁵ Incidentally, the figures in the two columns can not be added up to yield the total number of inventions patented by these different bodies, as some inventors sought and obtained a patent for the same invention both from the States General and from the States of Holland.⁶ The number of patents granted by other provincial estates and by town governments in the years between 1580 and 1720 was always much lower. The total number of patents issued in this period

Table 1 Number of patents granted by the States General and the States of Holland, 1580-1719

Period	Number of patents granted by the States General	Number of patents granted by the States of Holland
1580/89	9	9
1590/99	36	17
1600/09	55	8
1610/19	81	6
1620/29	118	1
1630/39	92	2
1640/49	39	24
1650/59	26	13
1660/69	38	36
1670/79	24	29
1680/89	7	28
1690/99	16	20
1700/09	7	22
1710/19	5	9

Source: Calculated on the basis of Doorman, *Octrooien*.

by the next most active province in the issue of patents, Utrecht, amounted to no more than fifteen.

Although patents could in principle be granted for inventions in every field of economic and technological activity, patented inventions were in reality throughout the period 1580-1719 largely concentrated in a limited number of fields, as Table 2 attests. The vast majority of patents granted by the States General and States of Holland related to inventions in the fields of hydraulic equipment and installations (water-raising mills, pumps, water supply systems, dredging equipment, bridges, sluices and the like), military technology (canon and shot, fortifications), heating equipment (kettles, furnaces, stoves and chimneys) and industrial windmill technology.

Between c.1580 and 1620 the patent system in the Republic became increasingly refined in the sense that provisions were introduced both to protect the interests of private inventors and to secure the interests of society at large. The interests of private inventors were guaranteed by the granting of exclusive rights to put their innovations into practice, the penalization of offenders, and the liberty to treat patents as alienable property which could be bought,

Table 2 Patented inventions in the Dutch Republic by field, c.1580-1719

Field	1580/ 1599 %	1600/ 1619 %	1620/ 1639 %	1640/ 1659 %	1660/ 1679 %	1680/ 1699 %	1700/ 1719 %
Hydraulic equipment and installations	45	32.4	38.0	31	29	21	40
Military technology	2	5.8	7.9	6	7	6	5
Navigation technology	2	0.6	—	1	1	2	5
Shipbuilding	2	2.5	0.3	2	1	6	8
Land transport	1	0.6	0.1	—	2	4	—
Heating equipment	10	7.1	4.8	2	8	2	5
Fire-fighting equipment	—	0.6	0.1	3	3	—	—
Precision instruments	2	1.2	0.1	2	6	—	3
Building equipment	1	2.5	0.3	—	3	10	—
Industrial windmills	23	13.6	18.1	13	8	4	3
Textile production	—	7.7	0.1	12	1	2	—
Ceramics	1	4.5	0.2	6	—	—	3
Iron and steel making	—	1.9	0.3	—	2	4	—
Food processing	1	1.2	0.2	1	1	—	3
Agricultural technology	—	1.2	0.2	2	2	—	3
Other	6	15.5	13.2	16	23	40	21
	100	100	100	100	100	100	100
N = Total number of patented inventions	79	154	226	85	96	52	37

Source: Calculated on the basis of Doorman, *Octrooien*.

sold, donated or inherited. Foreigners were treated on the same footing as native Dutchmen, on condition that (at least for a while) they had their actual residence in the Dutch Republic.⁷ Any benefit that might accrue from the legal recognition of the property right to an invention had to be reaped by the inventors themselves. In contrast with France, it was rare to find a public authority in the Dutch Republic support inventive activities by the grant of a premium or reward.⁸

The interests of society at large, on the other hand, were safeguarded by fixing a term during which the exclusive rights of patentees would apply (first usually five to twelve years, but after 1650 gradually settling on fifteen) and the obligation imposed on patentees to put their inventions into practice within a fixed period of time after the patent had been granted. From 1617 onwards, the length

of this period normally was put at one year.⁹ In common with a practice developed in Venice and various states in Germany, patentees in the United Provinces were moreover up to the 1620s in a number of cases explicitly obliged to grant a license to anyone who wished to make use of their invention and was willing to pay a reasonable price to do so.¹⁰ And both the interests of private inventors and of society at large were served by the prescription, introduced around 1590, that applicants for a patent had to submit a specification of their invention in the form of a drawing, model or description. Patentees themselves could use this specification as evidence to fight off competitors, whereas non-privileged parties could employ them to expose fraud.¹¹ Although patent disputes in the Dutch Republic were not a particularly frequent occurrence – the total number of cases from the period 1580–1720 that up to now has come to light, amounts to fifteen –, the mere possibility that a violation of a patent could be fought by taking legal action may have enticed at least a few inventors to prefer openness through patenting over secrecy as a strategy to protect their intellectual property.¹²

Before a patent was granted, both the application and the specification were normally examined by a committee composed of members from the States General or the provincial estates, who sometimes called on external experts for technical advice. Yet the inspection procedure was not as rigorous and thorough as sometimes has been thought.¹³ While the formal requirements for the obtainment of a patent in the Dutch Republic were more strict than in England before the 1730s, they were less exacting than those that obtained in eighteenth-century France. For one thing, it was not a standard practice that inventions submitted to the States General or provincial estates were subjected to scrutiny by a board of scientific experts. From c.1635 onwards, moreover, filing a specification of a patent became the exception rather than the rule; the practice did not revive until the 1770s. The reason for the lapse of the former prescription was probably, as Doorman has surmised,¹⁴ that the enrollment of a drawing, model or description was no longer deemed necessary when the new rule had come into force that patentees had to bring their inventions into practice within a year after the patent had been granted. Henceforth, the market would decide whether an invention was viable or not. It is thus going too far to claim, as Silberstein has done, that the decline in the number of patents granted by the States General in the eighteenth century was somehow related to a tightening up of the examination procedure.¹⁵

There is no evidence at all that the fall in the number of patents granted after 1700 coincided with, or was preceded by, a sharp rise in the rate of rejection of applications.¹⁶

In the foregoing I have used the term "society" rather than "state" to denote the other pole of the relationship that determined the evolution of patent law in the Netherlands in the early modern period. In contrast with France, central authorities in the Dutch Republic (or provincial estates, for that matter) did not pursue a deliberate policy of promoting the emergence and diffusion of inventions for the sake of increasing the wealth and power of the state. Although the States General and the States of Holland routinely showed an interest in any invention purported to enhance the capabilities of the Dutch armed forces on land or at sea, there was in the period under discussion not a concerted effort to harness inventive activity in the service of the Dutch in any systematic way. Inventors in the Dutch Republic never had to subscribe to an ethic of public service as nurtured in Louis XV's France.¹⁷

THE GEOGRAPHICAL ORIGINS OF PATENTEES

Of all inventors who succeeded in obtaining a patent from the States General or States of Holland between 1580 and 1720, the vast majority came from the Dutch Republic itself. As Table 3 demonstrates, the share of foreigners among those who called on the protection of the Dutch patent system decreased from the late sixteenth century until about 1660 and, after a brief upsurge in the sixties and seventies, stabilized at a fairly low level thereafter. Inside the Republic itself, the share of Holland was always much larger than that of all the other provinces combined, even in the time when most patents were still issued by the States General instead of by the estates of Holland. It was only in the period between c.1600 and 1640 that the outer provinces mustered more than 10% of those who received patents from either of these bodies. Considering the fact that Holland never held more than 42% of the total population of the Republic,¹⁸ this implied of course a strong over-representation of this province among the total number of patentees.

Inside Holland, the geographical spread of patentees showed a pattern that became more and more skewed as time wore on. Patenting grew increasingly dominated by the city of Amsterdam. As in England after 1660 or in eighteenth-century France,¹⁹ the share of

Table 3 Geographical origins of patentees in the Dutch Republic, 1580–1719

	1580/ 1599 %	1600/ 1619 %	1620/ 1639 %	1640/ 1659 %	1660/ 1679 %	1680/ 1699 %	1700/ 1719 %
Amsterdam	14	12.9	23.4	22	32	29	46
Other cities in Holland	42	31.0	34.3	41	22	16	33
Rest of Holland	10	14.2	4.0	9	7	3	
Rest of Dutch Republic	2	11.0	14.4	7	8	3	
Spanish Netherlands	6	2.6			1		
England	2	5.8	5.0	1	14	3	2
France		1.3	1.0			2	2
Germany	11	2.6	1.5	1	2		
Italy	2	2.6			1		
Other foreign countries		0.6				2	2
Unknown	13	14.2	16.4	19	14	42	15
	100	100	100	100	100	100	100
<i>N</i> = total number of known patentees	63	155	201	69	88	62	54

Source: Calculated on the basis of Doorman, *Octrooien*.

the metropolis among the holders of patents vastly exceeded its proportion of the population of the country at large. While the share of Amsterdam in the total population of the United Provinces increased from about 6% around 1620 to over 12% around 1720 (and in the total population of Holland from about 15% to over 30%),²⁰ its quota among patentees went up from about 30% to nearly 60% (and from about 38% to 60%, respectively). The growing predominance of Amsterdam among holders of patents is the more striking, as the middle decades of the seventeenth century saw an impressive improvement in the transportation system in the form of the completion of an extensive network of regular canal boat services between every town of importance in the maritime provinces of the Dutch Republic.²¹ In contrast to what Kenneth Sokoloff found in the United States in the first half of the nineteenth century,²² the extension of the transportation network in the Dutch case did not result in increased rates of patenting from those places that enjoyed improved access to a wider world. On the contrary, the distribution of patentees became much more narrow than before. Whereas in the period up to the 1640s almost every city in Holland could boast one

or two patentees per decade (and the small town of Alkmaar between 1600 and 1640 even spawned 23), there were by the beginning of the eighteenth century but a few of such centres of invention left. Besides Amsterdam, the only cities in Holland that still supplied a fair number of patentees around 1700 were Haarlem, Rotterdam and the town that served as the seat both of the States General and of the States of Holland, The Hague. The practice of patenting had in geographical respect contracted rather than diffused.

THE SOCIAL PROFILE OF PATENTEES

Trends in the status or occupational backgrounds of patentees in the Dutch Republic were not as drastic as changes in their geographical origins. The period between 1640 and 1720 saw only a minor shift in the ratio between artisans and the commercial and professional classes (merchants, lawyers, doctors/surgeons, military men, engineers/surveyors/architects and officials) with respect to the period up to 1640, when patenting reached its peak. While the share of the former group among patentees with known status or occupation fell from 53.6 to 46% (as Table 4 demonstrates), the proportion of the latter one rose from 30.5 to 40.2%. Among artisans, the share of masons and experts in wood-working, like carpenters, millwrights and shipwrights, declined in favour of those skilled in making instruments, such as watch-makers. Table 4 further shows that the representation of political elites among patentees on balance remained largely unchanged (even if the ratio between urban *regenten* and landed gentlemen was completely reversed).

Viewed from a comparative perspective, however, these data on occupations or status of patentees in the Dutch Republic nevertheless contain an interesting message. The figures shown in Table 4 are in stark contrast with the pattern that can be observed in England, the United States and France. In England, the proportion of "industrial producers" among patentees increased from 29% in the period 1660-1699 to 61% in the first half of the eighteenth century (and of artisans alone from 14 to 31%) and in the U.S. it rose from about 46% in the 1790s to 67% in the 1840s (and of artisans alone from 38 to 45%), while the share of the commercial and professional classes in these countries collapsed from over 50 to a mere 20%.²³ Although changes in France were not as spectacular as those in the Anglo-American orbit, there is in the middle of the eighteenth century

Table 4 Occupations or status of patentees in the Dutch Republic, 1580-1719

Occupation or status	1580-1639 %	1640-1719 %	1580-1719 %
Urban <i>regenten</i>	6.6	0.8	4.1
Landed gentlemen	4.0	11.0	7.0
Merchants	13.9	21.8	17.4
Lawyers	-	2.5	1.1
Doctors/surgeons	6.0	5.9	5.9
Military men	2.6	4.2	3.3
Engineers/surveyors/ Architects	7.3	4.2	5.9
Officials	0.7	1.6	1.1
Artisans			
Wood-workers	27.1	17.6	23.0
Metal-workers	10.6	10.9	10.7
Instrument makers	2.0	7.6	4.4
Masons	4.0	0.8	2.6
Pottery makers	0.7	1.6	1.1
Painters/carvers	1.3	4.2	2.6
Textile-makers	1.3	0.8	1.1
Others	6.6	2.5	5.4
Total	53.6	46.0	50.9
Farmers	1.3	-	0.7
Seafarers and fishermen	2.0	-	1.1
Others	2.0	-	1.8
N =	151 = 100%	119 = 100%	270 = 100%

Source: Calculated on the basis of Doorman, *Octrooien*.

among those who applied for a grant or a *privilege exclusif* from the *Conseil de commerce* still a rise in the share of artisans and *manufacturiers* to be noted from 59% to 72.5 (and of artisans alone from 33.6 to 56.1%).²⁴ The evolution of the social profile of patentees in the Dutch Republic between 1580 and 1720 thus differed from developments in other countries in the fact that the share of artisans already attained a relatively high level at a very early date, that the role of the commercial and professional classes increased rather than diminished and that 'industrial producers' other than artisans (such as centralized manufacturers) scarcely figured in the picture at all.

CAREERS IN PATENTING

As the numbers of patents issued by the States General and the States of Holland in the period after 1640 declined, so did the proportion of patentees who obtained more than one patent. In the period between 1580 and 1640, by contrast, "multiple" patentees were in the Dutch Republic a quite common appearance. According to the Dutch economic historian N.W. Posthumus "the inventor became a new and familiar type".²⁵ He singled out Simon Stevin (an engineer from Bruges who received patents for eleven inventions between 1584 and 1589), Willem Dircxz van Sonnevelt (a farmer from Voorschoten who received patents for four inventions between 1591 and 1606) and Cornelis Dircxz Zeeman (a carpenter and millwright from Warmont who received patents for five inventions 1627 and 1655) as pre-eminent examples. Although the proportion of "multiple" patentees of all patentees in this period never reached the level found for the U.S. in the first half of the 19th century (over 40%), it was exactly as high as the level in England between 1750 and 1800 (23%) and in all probability exceeded the level attained in eighteenth century France; the number of patentees in the Dutch Republic receiving four or more patents between 1580 and 1640 was higher than the total number of *inventeurs prolifiques* whom Hilaire-Perez identified in France between 1700 and 1789 (22 and 15, respectively).²⁶

But in the period after 1640, the ratio between "single patentees" and "multiple" patentees clearly underwent a change in favour of the former, as Table 5 attests. By the end of the seventeenth century

Table 5 Patentees in the Dutch Republic holding a single patent or more than one patent, as a proportion of all patentees, 1580-1719

Period	Single patentees %	Multiple patentees %	N
1580/99	73.2	26.8	46
1600/19	78.3	21.7	166
1620/39	78.3	21.7	198
1640/59	83.5	16.4	66
1660/79	86.7	13.3	90
1680/99	93.1	6.9	58
1700/19	83.3	14.7	48

Source: Calculated on the basis of Doorman, *Octrooien*.

there were hardly any individuals in the Dutch Republic left who could boast a kind of "career" in patenting. The upsurge after 1700 would prove to be short-lived.

It would be wrong, however, to equate these shifts in the relative importance of "multiple" patentees with the rise and decline of the "quasi-professional inventor" in the sense defined by Harry Dutton.²⁷ The extent of professionalization of "multiple" patentees in the Dutch Republic was, measured by Dutton's standards, certainly not high. Except in the period up to 1600, careers in patenting seldom ranged over a large number of inventions. The proportion of patentees who obtained four or more patents generally formed but a few per cent of the total number of people who received a patent from the States General or States of Holland. There was moreover not a strong propensity to diversify "inventive portfolio's" by inventing in a number of related sectors or industries. Most of the "multiple" patentees restricted their inventive activities to the trade in which they had first specialized. Carpenters and millwrights, for instance, usually concentrated their efforts on inventing or improving mills. But the fact that professionalization was still very weak did not preclude the presence of commercial tendencies in patenting. Even if "multiple" patentees only to a small extent displayed the features of "quasi-professional inventors", the Dutch Republic at an early date saw the rise of an entrepreneurial orientation among patentees and the rise of a trade in patented inventions.

COMMERCIAL TENDENCIES IN PATENTING

Commercial tendencies became manifest in several ways. The most elementary form in which they were expressed, was an agreement between a patentee and some other person to share the benefits of a newly-acquired patent. An early example was a deed drawn up before the town clerk's office of Delft in 1588, by which Simon Stevin stated that Johan de Groot, burgomaster of Delft, would henceforth enjoy equal rights to any profits that would accrue from the patent on a new water-raising mill which had been granted to Stevin in 1586.²⁸

Like in England in the eighteenth century, patentees in the Dutch Republic became involved in more complex forms of trading as well. As a successful exploitation of an invention by a patentee himself (in contrast to licensing) often required higher investments than the

patent-holder could afford, it often happened that inventors after the obtaining of a patent, or even in advance, sought financial support from outside sources. Christine MacLeod's observation that "partnership was a financial necessity forced on reluctant inventors" was valid for numerous inventors in the United Provinces as well.²⁹ In 1613, for instance, it was agreed by notarial deed that Gysbert Jansz Keyser, a mason from Amsterdam who had just received a patent from the States General on a new type of heating-device made from brick, and Jan Hendricksz Soop, the master of the local glass-works who was never mentioned in the patent at all but had apparently borne part of the expenses required to obtain it, would share the benefits and costs of the exploitation of Keyser's patent on a fifty-fifty basis.³⁰ In the same year, engineer Isaac Burger entered into a contract with merchant Andries van den Broeck which provided that Van den Broeck would furnish all the money needed to make fuel-saving iron stoves on a model recently patented by Burger and to enable the patentee to install furnaces in confectioneries, breweries, dye-houses and the like, in exchange for 50% and 33% of the profits, respectively. A dozen years later, Burger signed a similar agreement with another merchant concerning an invention for which he had not even yet secured a patent at all: an ice-breaking device to be used on rivers, canals and other waterways.³¹

Besides partnerships between a single inventor and a single "sponsor",³² there were also arrangements which involved a larger number of people and showed a higher degree of elaboration. When Burger in 1615 applied to the States General for a patent on a new method for reducing air pollution by burning coal, he teamed up with three associates, Thomas Pacque, Hendrick le Maire and Hans le Maire.³³ In 1626, Bruyning Hendricksz and Baltus Cornelisz agreed to admit Thomas Blanckaert as a "lifelong" partner in the exploitation of their newly obtained patent on a novel type of man-powered water-raising mill. The contract provided that Blanckaert would pay the patentees a lump sum of 500 guilders plus a fee of 600 guilders from the first profit the patent would yield. Hendricksz and Cornelisz meanwhile reserved the right to add a fourth partner if circumstances required.³⁴ In the same year, carpenter Jacob Sybrantsz formed a partnership with five associates – Philips and Guilbert de Flines, Paulus Buijs, Pieter van der Voort and Pieter Andriesz Moerbeek – to exploit a patent on a *trommelmolen* ("drum-shaped mill") for raising water of which he had recently finished a model. According to the elaborate arrangements laid down before a

notary in Amsterdam, Sybrantsz himself would keep 3/8 of the rights to the patent on this invention, while the remaining shares were transferred to his associates in the following proportions: Buijs, Van der Voort and Moerbeek 1/8 each and Philips and Guilbert de Flines together 2/8. None of the partners would be allowed to sell or alienate his share without permission from the others. In order to bring the invention into practice, each partner undertook to bear part of the costs of investments in proportion to his share in the patent (or risk losing his part of the profits). The actual building of the mill would be carried out by workmen under supervision of Sybrantsz, for a "reasonable" wage to be paid by his partners.³⁵

Sponsors could participate in more than one partnership at once. The role of merchant capitalists in the obtaining and exploitation of patents thus went further than the mere number of "merchants" among the holders of patents would suggest. The activities of merchant-brewer Pieter de Nys from Amsterdam form an interesting case in point. Although De Nys took only two patents in his own name (one of which concerned an invention made "by the ingenuity and practice" of somebody else),³⁶ he was in the 1620s and 1630s financially involved in the exploitation of at least three other ones. In 1622 and 1629 he contracted with a mason (Jacob Hutssen) and a blacksmith (Bartelt Cornelisz) to provide the money needed to bring their inventions concerning the pulverization of stone and the working of wrought iron into practice, in exchange for a certain share of the profit.³⁷ As an important creditor to Henric Frits and Abraham Jansz Segers, who held the rights to a patent on a water-raising mill invented by Segers and Pieter Sturck, De Nys in 1623 finally managed to get a large share in the exploitation of this patent as well.³⁸

The growth of the number of patents in the period up to about 1640 did not imply that patentees normally could take demand for their invention for granted. True, it now and then happened that an applicant for a patent beforehand could be sure that his invention would find a market. This was for instance the case, when an invention was made on request,³⁹ when a patent was applied for to protect the right to an invention that had already been put into practice⁴⁰ or when a patent was granted on the express condition that the holder of patent could license anyone to use his invention who wished to do so and was willing to pay for it.⁴¹ In the majority of cases, however, it was up to the patentees (or their associates) themselves to find a market for their new product, device or method.

For this purpose, patentees sometimes sought to stir publicity for their inventions through the staging of trials or by means of the spreading of flyers, pamphlets or books.⁴² A more usual way to enhance the chance of attracting a prospective buyer was probably, to seek out a concrete spot where a permanent interested public for inventions could be assumed to be found. This may have been one of the reasons, why the geographical spread of patentees in the seventeenth century became more and more skewed. Amsterdam, Haarlem or The Hague was a more suitable place to be for an ambitious inventor than, say, Alkmaar, Zwolle or Voorschoten. Finally, patentees could reduce the uncertainty of the market by contacting some institutional customer that could be expected to have both the interest and the means to procure one or more copies of their inventions. The first known applications of an improved water-raising mill patented by Simon Stevin and Johan de Groot in the late 1580s, for example, were made on behalf of the city government of Delft.⁴³ Given the fact that a large number of patents – as in the case of Stevin's and De Groot's mill – concerned inventions in the fields of hydraulic equipment and installations and military technology it was often the natural thing to do to seek potential buyers first and foremost among public or semi-public institutions like city governments, drainage boards, admiralties, army agencies or chartered companies like the Verenigde Oostindische Compagnie (VOC).

The variety of entrepreneurial strategies used by patentees in the United Provinces can in a nutshell be illustrated by looking at the case of the "multiple" patentee Jan van der Heyden. Van der Heyden, a painter by trade, who around 1650 moved from the small town of Gorcum to Amsterdam, received patents on fire-fighting equipment from the States General in 1672 and 1677, from the States of Holland in 1671 and 1677 and from the States of Friesland in 1678. The equipment consisted of a new type of suction pump to supply water to fire engines, a hose (made of leather and later of canvas) fitted to the engine itself and a pressure pump to push jets of water through the hose into the fire. In 1671 and 1672, Van der Heyden moreover obtained from the States of Holland and the States General a patent on a new kind of water-raising mill. In addition, during the late 1660s and 1670s he developed a number of inventions for which he never asked for a patent at all: a mud-mill, a stove and a new system for the lighting of cities. The latter consisted of a series of lampposts installed at a regular distance of *c.* 50 metres along streets and canals throughout the city, which were equipped

with square shaped tin lamps with glass panes, provided with air holes in the bottom and a chimney at the top. These lamps were, as usual, lit with oil.⁴⁴

Like many other inventors, Jan van der Heyden entered a sort of partnership to obtain and exploit his patents. The only difference was that his business partner was close kin. In 1672, he contracted before a notary with his younger brother Nicolaes to share the costs and benefits of the patents on the new fire-engine on a fifty-fifty basis.⁴⁵ But the most solid and enduring financial basis for his activities was provided by his long-term relationship with the government of Amsterdam. Following the decision of the burgomasters in 1669 to adopt his newly proposed system for city lighting, Van der Heyden received an appointment as supervisor-general of city lighting, which from 1670 until his death in 1712 yielded a steady income of 2000 guilders a year. When the city fathers in 1673 resolved to have part of the municipal fire-engines replaced by copies of the new type invented by Van der Heyden, Jan and his brother were engaged as supervisor-generals of the local fire-equipment as well, at a joint salary of 315 guilders a year.⁴⁶ By the early eighties, all fire equipment in the city was supplied by the workshop set up by the elder Van der Heyden.

The role of the city was also crucial for the very creation of these inventions. The inventiveness of Jan van der Heyden and his younger brother was spurred from the start by the knowledge that the government of Amsterdam was plainly in need of improved systems of city-lighting and fire-fighting. There was in their minds no doubt that inventions in these fields would find a ready market, because the city government had already considered similar proposals for the extension of city lighting shortly before and because the urban magistrate in 1672 had specifically asked Jan van der Heyden to inspect and if necessary repair all fire-extinguishing equipment in the city.⁴⁷ But the entrepreneurial ambitions of the elder Van der Heyden were not restricted to the city of Amsterdam alone. It was his declared aim to expand his market share in the sale of fire-fighting equipment in the Dutch Republic and beyond as far as he possibly could. Any infringement of his patents was firmly opposed. Between c.1680 and 1700 Jan van der Heyden conducted at least four patent suits. He did not hesitate to quarrel with his own kin about the division of the spoils.⁴⁸ He carried the publicity campaign for his own invention to greater lengths than any patentee in the Dutch Republic before, not only by publishing (in 1690) a full-fledged book with

drawings, statistics and "case histories" about his pathbreaking improvements in fire-fighting but also by practising a form of "direct mail" through sending copies of leaflets on his fire-engines to potential "institutional" customers abroad (like the city of Nuremberg).⁴⁹

And he was very succesful indeed. The new system of fire-fighting introduced by Van der Heyden in Amsterdam around 1670 was in the 1680s copied in other important cities in Holland like Gouda, Haarlem, Rotterdam, and Dordrecht. Fire-engines made by the Van der Heyden workshop were sold to widely different customers like the VOC, the Elector of the Palatine and the cities of Dresden and Cologne. No wonder that this patentee died as a well-to-do citizen. He was buried in the highest class, leaving an inheritance of nearly 84,000 guilders.⁵⁰

EPILOGUE: THE DECLINE OF PATENTING

The evolution of the patent system in the Dutch Republic thus reached a turning-point around 1640. While the number of patents granted by the States General and the States of Holland began a long-term decline and the filing of specifications became a more and more rare occurrence, the relative importance of "multiple" patentees diminished, the geographical spread of patentees lessened and the role of the commercial and professional classes – as distinct from manufacturers – increased. Except for the enhanced predominance of the metropolis among the holders of patents, the evolution of the system in the Dutch Republic after c.1640 tended to diverge from rather than converge to the pattern that has been observed for England in the latter half of the eighteenth century.

Although a decrease in the number of patents did not necessarily reflect a falling-off in the rate of invention, it certainly meant a decline in the demand for patents on the side of inventors. It would be going to far to ascribe this decline solely to the slowing down of the growth rate and eventual stagnation of the Dutch economy at large. The number of patents began to decrease even before the economy as a whole had reached the limits of its expansion. The downturn in the long-term development of the Dutch economy did not start until the middle of 1660s at the earliest.⁵¹ The growth of corporatism does not provide an adequate explanation for the decline in the demand for patents on the side of inventors either.⁵² True, the number of guilds in the Dutch Republic rapidly rose in the course of

the seventeenth century and especially during the period 1600–1650. By the end of the seventeenth century, a much larger part of the urban economies in United Provinces was dominated by corporate institutions than in the 1580s or 1590s.⁵³ But the expansion of “distributional coalitions” can not have substantially altered the overall possibilities for individual inventors to seek protection of intellectual property by applying for a patent from the States General or provincial estates, as the very fields that attracted a relatively large number of patents – such as hydraulic equipment and installations, military technology, heating equipment and industrial windmills – still largely remained outside the domain of such corporate organizations.

I would venture that the decline in the demand for patents on the side of inventors must at least in part have been due to the evolution of the patent system itself. As the decision about the viability of a patented invention from the 1630s onwards was in fact increasingly left to the market, while all patentees were put under an obligation to bring their invention into practice within a year after the grant of the patent, the benefits of obtaining a patent at all were from the point of view of inventors probably becoming less clear. In these circumstances, the chances to reap the benefits offered by a patent would have been highest if an inventor resided close to a spot where a permanent interested public for inventions could be assumed to be found (for example in Amsterdam, Haarlem or The Hague) or if customers could be found among institutions with a considerable amount of money to spend like a city government, a chartered company or an army agency. The career of Jan van der Heyden is a case in point.

The decline in the public offer of inventions that resulted from the changes in the patent system would perhaps to some extent have been compensated if the States General or provincial estates, like state agencies in France, had stimulated inventors to come forward by granting rewards or premiums on a regular basis. In the end, it was not the public authorities of the Republic but a private organization, the *Oeconomische Tak van de Hollandsche Maatschappij der Wetenschappen*, founded in 1777 with the aim of promoting the general wealth of the Dutch nation and its colonies by the diffusion of useful knowledge in every branch of economic life, which first introduced such a system of material incentives into the Dutch Republic.⁵⁴ The significance of the patent system rapidly paled by comparison. While the total number of patents granted by the States General and the

States of Holland in the period 1778–1797 came to no more than 23 and three, respectively, the number of rewards and premiums paid by the *Oeconomische Tak* between 1778 and 1787 amounted to 282 and in the next decade to more than 90!⁵⁵ The patent system in the Dutch Republic was by then apparently lacking in sufficient incentives.

Notes

1. G. Doorman, *Octrooien voor uitvindingen in de Nederlanden uit de 16e–18e eeuw* (The Hague 1940), *Patent Law in the Netherlands* (The Hague, 1942), M. Silberstein, *Erfindungsschutz und merkantilistische Gewerbeprivilegien* (Winterthur, 1961).
2. Silberstein, *Erfindungsschutz*, p. 253.
3. An outline of technological developments in the Dutch Republic is given in the following articles I have published in recent years: "Technological Change and the Economic Expansion of the Dutch Republic, c.1580–1680", in: Karel Davids and Leo Noordegraaf (eds), *The Dutch Economy in the Golden Age. Nine Studies* (Amsterdam, 1993), pp. 79–104; "Shifts of Technological Leadership in Early Modern Europe", in: Karel Davids and Jan Lucassen (eds), *A Miracle Mirrored. The Dutch Republic in European Perspective* (Cambridge, 1995), pp. 338–366; "Openness or Secrecy? Industrial Espionage in the Dutch Republic", *The Journal of European Economic History* (1995) 24: 333–348; "Successful and Failed Transitions. A Comparison of Windmill-Technology in Britain and the Netherlands in the Early Modern Period", *History and Technology* (1997) 14: 1–23.
4. Christine MacLeod, *Inventing the Industrial Revolution. The English Patent System, 1660–1800* (Cambridge, 1988); B. Zorina Kahn, "Property Rights and Patent Litigation in Early Nineteenth Century America", *Journal of Economic History* (1995) 55: 58–97; B. Zorina Kahn and Kenneth L. Sokoloff, "'Schemes of Practical Utility'. Entrepreneurship and Innovation among 'Great Inventors' in the United States 1790–1865", *Journal of Economic History* (1993) 53: 289–307; Kenneth L. Sokoloff, "Inventive Activity in Early Industrial America: Evidence from Patent Records", *Journal of Economic History* (1988) 48: 813–850; Kenneth L. Sokoloff and B. Zorina Kahn, "The Democratization of Invention during early Industrialization: Evidence from the United States, 1790–1845", *Journal of Economic History* (1993) 50: 363–378; Richard J. Sullivan, "The Revolution of Ideas: Widespread Patenting and Invention during the English Industrial Revolution", *Journal of Economic History* (1993) 50: 349–362; Liliane Hilaire-Perez, *Inventions et inventeurs en France et en Angleterre au XVIIIe siècle* 4 vols. doctorat Université de Lille, Paris I, January 1994. Cf. also the wide-ranging overview by Paul A. David, "Intellectual Property Institutions and the Panda's Thumb: Patents, Copyrights, and Trade Secrets in Economic Theory and History", in: Michael B. Wallerstein, Mary Ellen Mogege, Roberta A. Schoen (eds), *The Global Dimensions of Intellectual Property Rights in Science and Technology* (Washington, 1993), pp. 19–61.
5. Paul Hofstijzer, "Nederlandse boekverkopersprivileges in de zeventiende en achttiende eeuw", *Jaarboek van het Nederlandse Genootschap van Bibliotheekwetenschappers* 1993: 49–62 on p. 55.
6. Cf. Silberstein, *Erfindungsschutz*, p. 85.
7. Doorman, *Octrooien*, p. 21.
8. Doorman, *Octrooien*, pp. 29–30; C.A. Davids, *Zeezeven en wetenschap. De wetenschap en de ontwikkeling van de navigatietechniek in Nederland tussen 1585 en 1815* (Amsterdam/Dieren, 1986), p. 82.

9. Doorman, *Octrooien*, pp. 20–28; N.W. Posthumus, *De geschiedenis van de Leidsche lakenindustrie* 3 vols. (The Hague, 1908–1939), II, p. 395.
10. Silberstein, *Erfindungsschutz*, p. 21, 86; examples in Doorman, *Octrooien*, pp. 96, 117, 119, 121, 123, 125.
11. Doorman, *Octrooien*, pp. 21–22; J. Coren, *Observationes rerum in senatu iudicatorum in item consilia quaedam* (The Hague, 1633) pp. 159–264, especially pp. 161–162.
12. Examples of contracts to keep inventions secret: J.G. van Dillen (ed.) *Bronnen tot de geschiedenis van het bedrijfsleven en gildewezen van Amsterdam*, 3 vols. (The Hague, 1929–1951), I, nr. 1191 contract 18 april 1611, III, pp. 738–739 nr. 1557 contract 19 January 1666, Gemeentearchief Amsterdam (GAA), Notarieel Archief (NA) 130 f. 198–199 contract 28 March 1613, 841 contract 19 January 1629; examples of difficulties of keeping inventions secret: Algemeen Rijksarchief (ARA), Archief Staten Generaal (SG) 12301 f. 222 patent 6 August 1615, Van Dillen (ed.), *Bronnen*, II, p. 150 nr. 270 contract 21 March 1616, nr. 340 contract 17 November 1616, pp. 234–235 nr. 378 insinuation 21 April 1614.
13. MacLeod, *Inventing the Industrial Revolution*, p. 41, following Doorman, *Octrooien*, p. 20.
14. Doorman, *Octrooien*, pp. 22, 24–25.
15. Silberstein, *Erfindungsschutz*, p. 105; a permanent committee of the States General for the examination of inventions was already in operation in the 1650s, and not only from around 1700 onwards, as Silberstein supposed, see G. de Bruijn, *Geheimhouding en verraad. De geheimhouding van staatszaken ten tijde van de Republiek (1600–1750)* (The Hague, 1991), pp. 141–142.
16. The rate of rejection of applications by the States General only underwent a sharp rise after 1730 and the rate of rejection by the States of Holland remained low even after that date, see the data in Doorman, *Octrooien*, pp. 260–271, 308–320.
17. Liliane Hilaire-Pérez, *Inventions et inventeurs en France et en Angleterre au XVIIIe siècle* 4 dln. doctorat Université de Lille, Paris I, janvier 1994, t. 1, 58–242; Liliane Hilaire-Pérez, "Invention, politique et société en France dans la deuxième moitié du XVIIIe siècle", *Revue d'histoire moderne et contemporaine* (1990) 37: 36–63.
18. Jan de Vries and Ad van der Woude, *Nederland 1500–1815. De eerste ronde van moderne economische groei* (Amsterdam, 1995), pp. 71–79.
19. MacLeod, *Inventing the Industrial Revolution*, pp. 118–119; Hilaire-Pérez, *Inventions*, t. 4, annexe 6 and 8.
20. Calculated on the basis of figures from Marco van Leeuwen and James E. Oepen, "Reconstructing the Demographical Regime of Amsterdam 1681–1920", *Economic and Social History in the Netherlands* (1993) 5: 61–102 and De Vries and Van der Woude, *Nederland 1500–1815*, p. 74.
21. Jan de Vries, *The Dutch Rural Economy in the Golden Age 1500–1700* (New Haven, 1974), pp. 207–208 and Barges and *Capitalism. Passenger Transportation in the Dutch Economy (1632–1839)* (Wageningen, 1978, Utrecht³).
22. Sokoloff, "Inventive Activity", pp. 830–843.
23. MacLeod, *Inventing the Industrial Revolution*, pp. 134–135; Sokoloff, "Democratization", p. 369.
24. Hilaire-Pérez, *Inventions*, t. 4, annexe 8.
25. Posthumus, *Leidsche lakenindustrie*, II, p. 396.
26. Sokoloff, "Democratization", p. 367; MacLeod, *Inventing the Industrial Revolution*, p. 140; Hilaire-Pérez, *Inventions*, t. 2, pp. 25–26.
27. H.I. Dutton, *The Patent System and Inventive Activity during the Industrial Revolution* (Manchester, 1984), p. 108; the following findings are consistent with the results of MacLeod's careful analysis of the data on English patentees in the period 1750–1799, MacLeod, *Inventing the Industrial Revolution*, pp. 139–141.
28. G. Doorman, *Octrooien voor uitvindingen in de Nederlanden uit de 16e–18e eeuw. Eerste aanvulling* (The Hague, 1942), p. 19.

29. MacLeod, *Inventing the Industrial Revolution*, p. 90.
30. Van Dillen (ed.), *Bronnen*, II, pp. 56–57, nr. 104 contract 16 August 1613.
31. Van Dillen (ed.), *Bronnen*, II, pp. 32–33 nr. 62 contract 7 March 1613, Doorman, *Octrooien*, p. 123; GAA NA 841 f. 238–240 contract 12 December 1626; in the end, Burger has never applied for a patent on this invention at all. Other examples of similar contracts in: Van Dillen (ed.), I, pp. 700–703 nr. 1186 contract 31 December 1610, II, p. 257 nr. 412 contract 2 November 1617, pp. 305–307 nr. 501 21 September 1618, GAA NA 730 B f. 705 contract 30 June 1638, NA 522 f. 208 notarial deed 27 April 1638, NA 137 f. 574–581 contract 21 June 1652.
32. Other examples in: Van Dillen (ed.), *Bronnen*, II, 141 nr. 258 17 January 1616, 174–176 nr. 320 contract 23 August 1618.
33. Doorman, *Octrooien*, pp. 129–130.
34. Doorman, *Octrooien*, p. 168; GAA NA 845 contract 21 September 1626.
35. Doorman, *Octrooien*, p. 169; GAA NA 440 f. 226–227v, NA 468 f. 602 contract 9 december 1626.
36. Doorman, *Octrooien*, pp. 172, 183–184.
37. Van Dillen (ed.), *Bronnen*, II, pp. 452–453 nr. 781 contract 10 June 1622, II, p. 676 nr. 1212 contract 20 May 1629.
38. Doorman, *Octrooien*, pp. 149–150; GAA NA 716 f. 665 contract 16 June 1622, f. 763 contract 13 August 1622; Van Dillen (ed.), *Bronnen*, II, pp. 436–437 nr. 752 contract 12 February 1622, p. 454 nr. 786 contract 17 June 1622, pp. 489–490 nr. 854 contract 22 April 1623.
39. Claes Wassenaeer, *Historische verhael alder ghedenckwaerdichste geschiedenisse*, vol. 8, 86^v on an invention by Jan Osborn at the behest of the Noordse Compagnie.
40. ARA SG 12301 f. 222 patent 6 August 1615.
41. See note 10.
42. A fine example is: *Wercking van het geoctroyeerde water-scheprad geïnventeert by William Wheeler* (Amsterdam, 1649); the appointment by Simon Hulsebos of an agent to protect his interests as a patentee was an exceptional case, see GAA NA 733 B notarial deed 3 October 1644.
43. Doorman, *Octrooien. Eerste aanvulling*, pp. 19–20.
44. Doorman, *Octrooien*, pp. 240, 245, 296, 300–301, 326; Robert Multhauf, "The Light of Lamp-lanterns: Street Lighting in 17th century Amsterdam", *Technology and Culture* (1985) 26: 236–252; J.C. Breen, "Jan van der Heyden", *Jaarboek Amstelodamum* (1913) 11: 29–92 esp. pp. 42–52.
45. A. Bredius, "De nalatenschap van Jan van der Heyden's weduwe", *Oud Holland* (1912) 30: pp. 129–151 on p. 139.
46. Breen, "Jan van der Heyden", pp. 45–47, 50–55.
47. Breen, "Jan van der Heyden", p. 49. A request from the city government was in itself not a sufficient condition for the actual adoption of an invention, however, as the cases of the mud-mill and the stove prove. Although these inventions were made at the behest of the city fathers as well, they were never adopted, see Breen, "Jan van der Heyden", p. 48.
48. Breen, "Jan van der Heyden", pp. 53–55, 58.
49. Jan and Nicolaes van der Heyden, *Beschryving der nieuwljcks uitgevonden en geoctroyeerde slang-brandspuiten en hare wijze van brand blussen tegenwoordig binnen Amsterdam in gebruik zijnde* (Amsterdam, 1690); Stadtarchiv Nürnberg, Akten Reichstädtisches Bauamt, XXIII, 18.
50. H. Wildeboer, "De ontwikkeling van de brandspuit in de zeventiende en achttiende eeuw", *Jaarboek voor de Geschiedenis van Bedrijf en Techniek* (1993) 10: pp. 9–41 on pp. 33–35; Breen, "Van der Heyden", pp. 60–61; Bredius, "De nalatenschap", p. 130;
51. De Vries and Van der Woude, *Nederland 1500–1815*, chapter 13.

52. Cf. Silberstein' thesis in *Erfindungsschutz*, p. 85: "Monopolpatente (wurden) wahrscheinlich nur verliehen (...), wenn es sich um Produkte des freien, nichtzünftigen Gewerbe handelte".
53. Jan Lucassen and Piet Lourens, "Ambachtsgilden in Nederland: een eerste inventarisatie", *NEHA-Jaarboek voor de economische, bedrijfs - en techniekgeschiedenis* (1994) 57: pp. 34-62 esp. pp. 42-48.
54. J. Bierens de Haan, *Van Oeconomische Tak tot Nederlandsche Maatschappij voor Nijverheid en Handel 1777-1952* (Haarlem, 1952) pp. 1-15, 43-53; like the Société d'Émulation de l'abbé Baudeau in France, this organization was modelled on the Society for the Encouragement of Arts, Manufacture and Commerce in London.
55. Doorman, *Octrooien*, pp. 268-271, 316-320; S. van der Paauw, *Algemeen register van de prijsvragen, medailles en premien uitgeschreven en toegekend door de Nederlandse Maatschappij ter bevordering van Nijverheid te Haarlem sedert hare oprichting* (Haarlem, 1849) pp. 71-106.